



Google's current complete build of its driverless car

1. Limited autonomy that doesn't depend on data sharing:

This is the stage that we are at right now. Most vehicles now have basic sensors like cameras, proximity sensors to warn the driver, lane keeping/lane warning systems and tragic blind spot detection system. Even the smart Autopilot technology by Tesla in its cars and is actually a combination of all of the above and is a part of this basic stage. These technologies are powerful and form the basis of transition to self driving cars.

2. Semi-autonomy that utilizes shared data:

This stage is the actual beginning where the market penetration can start to happen and is about 5-10 years away from the present. The major part in this stage is V2X (Vehicle to 'X') communication. This is nothing but the communication between the vehicle and 'X' which can either be other vehicles or the surroundings or infrastructure around the vehicle. This would allow the vehicle to perceive the world around it as it actually is using 3D models. This is how autonomous vehicles can bridge the gap between available information from other vehicles and the road itself. At this stage, at least 50% of all vehicles on road must have IoT based sensing and connectivity systems in order to move forward.

3. Full autonomy with shared data, advanced sensors and high resolution mapping:

In order to achieve complete autonomy, this stage should have a combination